

CIE – 2 Syllabus (Deep Learning)

1. Introduction to Deep Learning and Neural Networks

- **Deep Learning:** A branch of machine learning where models learn features automatically from raw data using multiple neural network layers.
 - **Neural Network:** Inspired by the human brain, it consists of layers of “neurons” that process input and give output.
 - **Key Idea:** Each layer learns some feature. More layers → more complex patterns.
 - **Example:**
 - Shallow Neural Network → Detects edges in an image.
 - Deep Neural Network → Detects faces, emotions, and even objects.
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2. Fundamentals of Neural Networks

- **Components:**
 - **Input Layer** → Takes input data (e.g., pixels of an image).
 - **Hidden Layers** → Process and transform input using weights & activation functions.
 - **Output Layer** → Gives the prediction result.
- **Weights & Bias:** Parameters that control learning.
- **Activation Functions:** Decide whether a neuron should be activated (ReLU, Sigmoid, Tanh).
- **Example:** Predicting whether an email is spam or not using a neural network.

3. Feedforward Neural Networks (FNNs) and Backpropagation

- **FNN:** Information flows only in one direction (Input → Hidden → Output).
- **Backpropagation:** Learning algorithm that adjusts weights using error signals.
- **Steps:**
 1. Forward Pass (prediction).
 2. Calculate Error (difference between prediction & actual).
 3. Backpropagation (adjust weights using gradient descent).
- **Example:** Handwritten digit recognition (MNIST dataset).

4. Recurrent Neural Networks (RNNs)

- **Definition:** Neural networks where output depends not only on current input but also on past inputs (sequence learning).
- **Special Feature:** "Memory" through recurrent connections.
- **Applications:** Text, speech, time series.
- **Example:** Predicting the next word in a sentence ("I am going to the ____").

5. Convolutional Neural Networks (CNNs)

- **Definition:** Neural networks specialized for image and video data.
- **Key Concepts:**
 - **Convolution Layer** → Extracts features using filters (edges, shapes).
 - **Pooling Layer** → Reduces size but keeps important info.

- **Fully Connected Layer** → Final prediction.
 - **Applications:** Image classification, object detection, face recognition.
 - **Example:** Google Photos automatically recognizing a dog in your pictures.
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6. Generative Adversarial Networks (GANs) and Autoencoders

- **GANs:** Two networks (Generator + Discriminator) compete.
 - Generator → Creates fake data.
 - Discriminator → Detects if data is real or fake.
 - **Example:** Creating realistic fake human faces (deepfakes).
 - **Autoencoders:** Neural networks that compress data (encoder) and reconstruct it (decoder).
 - **Example:** Noise removal from images or dimensionality reduction.
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7. Reinforcement Learning (RL)

- **Definition:** An agent learns by interacting with an environment and receiving rewards/punishments.
- **Key Concepts:**
 - Agent, Environment, Reward, Policy.
 - Goal: Maximize long-term reward.
- **Example:**
 - Teaching a robot to walk.
 - AlphaGo beating humans in the game of Go.

8. Pre-trained Models and Fine-tuning

- **Pre-trained Models:** Models trained on large datasets (like ImageNet) and shared publicly.
- **Fine-tuning:** Reusing a pre-trained model and training it further for a new, smaller task.
- **Benefit:** Saves time, computation, and data.
- **Example:** Using a pre-trained CNN (ResNet) for detecting diseases in medical X-rays.